SC4001 Neural Networks and Deep Learning
Group Project
GROUP NUMBER: GROUP 127
Topic F: Retrieval Augmented Generation

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AI Use Statement

Claude Code (Anthropic, Claude Opus 5.5) used to help with the idea formulation for the E1 method to tackle pain points encountered in E0. We use the same AI to assist in the codes for E3 and E4 notebooks and their README documentation, and to edit this report. The report was checked against the notebooks' printed outputs and result files, and the references were checked against the original publications. Our team chose the research question, designed the experiments and interpreted the results and reviewed the codes written by agents.

1.Introduction

Retrieval-augmented generation (RAG) answers a question by giving a language model a set of retrieved documents and asking it to answer from them. The approach reduces reliance on the model's parametric memory, but it moves the reliability problem to the evidence. Retrieved passages are often off-topic, sometimes none of them contain the answer, and occasionally two passages disagree. A reliable system should answer correctly when the evidence supports an answer, refuse when it does not, and avoid repeating false information.

This project studies these three failure conditions with a small, openly available model, Qwen2.5-3B-Instruct, on the English part of the RGB benchmark (Chen et al., 2024). Our research question is: ***For Qwen2.5-3B-Instruct on RGB, how do passage-relevance filtering and score-based abstention, separately and together, affect answer accuracy and rejection of unanswerable questions?***

We built a prompt-based baseline (E0), two families of simple reliability methods (E1: embedding-similarity filtering with similarity-based abstention), and a method that scores every passage with the model itself (E2). E2 has two parts: a filter that removes passages judged irrelevant, and an abstention rule that refuses when even the best passage scores low. Because E2 changes two things at once, a single comparison against E0 cannot say which part helped. Our main contributions are therefore an ablation (E3) that separates the two parts with a 2x2 design, and a question-level failure analysis (E4) that explains where the remaining errors come from.

The main findings are that filtering improves the balanced score mainly through increased rejection of unanswerable questions, while additional score-based abstention provides no further benefit in this setting. Most remaining failures occur despite retained gold evidence, and relevance filtering alone does not resolve contradictory passages.

2.Review of existing techniques

Retrieval-augmented generation. Lewis et al. (2020) combined a dense retriever with a sequence-to-sequence generator and showed gains on knowledge-intensive tasks. Later work found that models are sensitive to what is retrieved. Shi et al. (2023) showed that adding irrelevant context lowers accuracy on reasoning problems, and Liu et al. (2024) showed that models use information at the start and end of a long context better than information in the middle. These results motivate removing irrelevant passages rather than passing everything to the model.

Benchmarks for RAG robustness. RGB (Chen et al., 2024) evaluates four abilities: **noise robustness, negative rejection, information integration and counterfactual robustness**. It supplies, for each question, passages labelled as containing the answer (positive) or not (negative), and for counterfactual questions a copy of the positive passage edited to state a false answer. These labels let us measure passage selection directly, which is why we chose it. We use the first two abilities and counterfactual robustness.

Evidence selection. A common approach re-ranks retrieved passages with a stronger model. Bi-encoders such as Sentence-BERT (Reimers and Gurevych, 2019) embed the question and passage separately and compare them by cosine similarity; they are fast but mainly measure topical similarity. Cross-encoders score the question and passage jointly (Nogueira and Cho, 2019) and are more accurate. Yoran et al. (2024) used a natural-language-inference model to discard irrelevant passages, and Corrective RAG (Yan et al., 2024) trains a lightweight evaluator that grades retrieved documents and triggers different actions depending on the grade. Self-RAG (Asai et al., 2024) trains the generator itself to emit reflection tokens that judge whether a passage is relevant and whether the answer is supported. Our E2 is a training-free version of this idea: the same 3B model is prompted to judge each passage, and its probability of answering "Yes" is used as a relevance score.

Abstention and uncertainty. A model can be made to refuse by instruction, as in the RGB prompt, or by thresholding a confidence signal. Kadavath et al. (2022) showed that larger models are reasonably calibrated when asked whether they know an answer. Sampling-based signals measure agreement between several sampled answers (Manakul et al., 2023; Kuhn et al., 2023). Our E1 baseline abstains on the embedding similarity between the question and its best passage; E2 abstains on the generator's own judgement of passage relevance. Both decide before the answer is generated.

The gap we address is attribution. Methods that both filter and abstain are usually reported as one system, so it is unclear whether a gain comes from better evidence or from refusing more often, which inflates any metric that rewards refusals. We test this directly.

3.Experimental Setup and Methods

A) Data and evaluation conditions

We use two RGB files: *en_refine.json* (300 questions, the authors' corrected version of the English set) and *en_fact.json* (100 counterfactual questions). Each question receives five passages, shuffled with a fixed seed so that the gold passage's position carries no signal. A few questions have only three or four passages because RGB supplies fewer (1,293 contexts have five, three have four and four have three). We build five conditions:

- noise_0.0, noise_0.4, noise_0.8: a fraction *r* of the five passages are negatives and the rest are positives, so the answer is always present.
- incomplete: all five passages are negatives (*r = 1.0*); the correct behaviour is to refuse.
- contradictory: one true positive passage, one counterfactual passage stating a fake answer, and three negatives.

This gives 1,300 contexts. Questions are split by ID with seed 42 into a 30% calibration split and a 70% test split, using the same split for every method and condition. All thresholds are chosen on the calibration split only; every number reported below is on the test split of 910 contexts (210 per noise and incomplete condition, 70 contradictory). The exact contexts are exported with a SHA-256 hash, and every later

notebook checks the hash before using them.

B) Generator and prompt

The generator we used is Qwen2.5-3B-Instruct (Qwen Team, 2024), 4-bit quantized (Q4_K_M) and served through Ollama, with greedy decoding (temperature 0, seed 42, at most 64 new tokens). Each call sends the following system and user messages; passages are numbered in shuffled order with whitespace collapsed, which is our [E0] prompt-based refusal baseline:

> **System:** *You are an accurate and reliable assistant that answers questions using only the provided documents.*
> **User:**
> *Documents:*
> *[1] {passage_1} ... [5] {passage_5}*
> *Question: {question}*
> *Instructions:*
> *- If the documents contain the answer, reply with a short answer.*
> *- If the documents do not contain enough information to answer, reply exactly: "I can not answer the question because of insufficient information in documents."*
> *- If the documents contain conflicting or incorrect information, start your reply with "There are factual errors in the provided documents." and then give the correct answer.*

The three instructions follow RGB's noise-robustness, negative-rejection and counterfactual settings. All methods use this template and differ only in which passages are listed, and E1 and E2 output the same refusal sentence when they abstain. E0 and E1 was generated on an Ollama server on the NTU TC1 cluster (one V100 GPU).

C) Scoring

We follow RGB's answer matching: a prediction is correct if it contains the gold answer as a case-insensitive substring, where the answer may require several parts and each part may have alternatives. A prediction counts as a refusal if it contains the refusal sentence or a common refusal phrase in the model's own words, unless it also contains a hedge marker such as "likely" or "probably" ("not explicitly stated, but it is likely X" is treated as an answer). Every answerable prediction is labelled *correct*, *rejected* (refused although the answer was available) or *wrong*; every incomplete prediction is *refused* or *answered*.

Our selection metric is the balanced score:

$$\textit{balanced} = \tfrac{1}{2} \times (\textit{mean accuracy over noise_0.0, noise_0.4 and noise_0.8} + \textit{incomplete rejection rate})$$

It weighs answering correctly and refusing correctly equally, so a method cannot score well by always answering or always refusing. Contradictory questions are reported separately and never used for threshold selection, because the desired behaviour there (answer correctly and flag the conflict) is different.

D) Methods compared

E0, prompt-based refusal baseline. The model sees all passages and relies on the prompt to refuse.

E1, similarity filter and abstention baseline. Passages are embedded with all-mpnet-base-v2, the dense retriever RGB used to collect them. The two passages most similar to the question are kept, and the system abstains when the best cosine similarity is below a threshold τ chosen on the calibration split.

E2, Qwen passage-relevance filter with abstention. For every passage, the same Qwen model is asked "Does this passage contain information that answers the question?" with a system prompt requiring a Yes or No answer. From the top-20 log-probabilities of the first generated token we compute s = P(Yes) / (P(Yes) + P(No)). Two thresholds are then chosen on the calibration split:

- The filter threshold maximises passage-level F1 for identifying RGB's positive passages in the three noise conditions (calibration F1 0.875 over 1,350 passages; threshold 1.37×10^{-7}). At test time, passages with s at or above the threshold are kept in their original order. If none qualify, the single highest-scoring passage is kept (the fallback).
- The abstention threshold τ maximises the calibration balanced score when the answer is replaced by the refusal sentence for questions whose best original-passage score is below τ ($\tau = 4.80 \times 10^{-7}$). It was chosen using E0's answers.

The scores are very small because the model is confident in most of its "No" judgements; only their order matters for the thresholds.

E3: ablation of filtering and abstention

Abstention depends only on the passage scores, not on the generated answer, so it can be applied to saved predictions without generating again. This makes an exact 2x2 ablation possible from two generation runs. We rebuilt four test-split variants: E0 (no filter, no abstention), E2 filter only, E0 abstain only (all passages, then the score rule) and E2 full (filter and abstention). Before analysing them, the notebook checks the context hash, the split sizes and score alignment, and reproduces E2's reported E0 and E2-full metrics to within 10^{-9}. For each pair of variants it counts label transitions (for example correct to rejected) and checks that they add up to the accuracy differences. It then asks whether abstention was simply mistuned: τ is chosen again on the calibration split using the filter-only answers (τ'), and a sweep over 60 log-spaced thresholds on the test split gives a descriptive threshold-sensitivity analysis.

E4: question-level failure analysis

For every test question we record the passage scores, which passages were kept, whether the fallback was used, whether each passage contains the gold answer, and the four predictions and labels. The gold-evidence flag uses the same substring match as the answer metric; it agrees with RGB's labels, since 99.8% of positive passages and 0.7% of negative passages contain the gold string. For contradictory questions, the true and fake passages are identified exactly by text membership in RGB's *positive* and *positive_wrong* fields. Each failure of the full E2 system is then assigned one category by fixed rules, applied in this order:

- abstention_override: the filter-only answer was correct, but the score rule replaced it with a refusal.
- filter_removed_gold: a passage containing the gold answer existed but the filter removed all such passages.
- gold_not_found: no passage contains the gold string.
- refused_despite_evidence / wrong_despite_evidence: a gold passage was kept, but the model refused or answered wrongly.
- answered_without_evidence: an incomplete question that was answered.

The notebook rebuilds the variants independently and asserts that its scores and transitions match E3 exactly before producing these tables.

4.Experiments and results

A) Baselines and E2 comparison

Table 1 reports the E0 and E1 baselines against E2 on the numeric-ID 30/70 split. The baselines are the same Ollama run on different machines, so every result in this report comes from one model configuration.

Table 1. Baseline and E2 results on the same test questions. Acc = accuracy; rej = rejection rate; Bal. = balanced score

Method	Acc 0.0	Acc 0.4	Acc 0.8	Rej incomplete	Acc contra	Rej contra	Balanced
E0 prompt refusal	0.948	0.971	0.719	0.852	0.543	0.314	0.866
E1 MPNet top-2 + abstain (τ = 0.447)	0.886	0.838	0.700	0.929	0.329	0.471	0.868
E2 Qwen filter + abstain (Ollama)	0.943	0.929	0.700	0.952	0.343	0.529	0.905

E1 raises incomplete rejection from 0.852 to 0.929 but loses answerable accuracy in every noise condition (6.2, 13.3 and 1.9 points at noise 0.0, 0.4 and 0.8). Its balanced score (0.868) is therefore almost the same as E0's (0.866), a difference of roughly one question. The loss at noise_0.0 cannot come from discarding the answer, because every passage there is positive; it comes from abstention or from the model answering differently when it sees only two passages. E1 also lowers contradictory accuracy from 0.543 to 0.329 and raises contradictory rejection from 0.314 to 0.471. A fixed top-2 similarity cut thus trades answers for refusals, which motivates a filter that keeps a variable number of passages according to judged relevance.

Full E2 scores 0.905, against 0.866 for E0 and 0.868 for E1. It rejects the most incomplete questions (0.952, against 0.852 and 0.929) and keeps more answerable accuracy than E1 at noise 0.0 and 0.4 (0.943 and 0.929, against 0.886 and 0.838), while matching it at noise_0.8 (0.700). Its noise_0.8 accuracy is still below E0 (0.719), and its contradictory accuracy (0.343) is well below E0 (0.543) and close to E1 (0.329). All three methods share the same model, prompt and decoding settings, so these differences reflect passage selection and abstention rather than the inference setup.

At noise 0.0, 0.4, and 0.8, the filter retains on average 89.0%, 91.0%, and 95.7% of positive passages, with mean passage precision 1.00, 0.89, and 0.67. It keeps approximately 4.5, 3.2, and 1.8 passages respectively. On incomplete questions, 93 of 210 use the single-passage fallback, and the mean number kept is 1.48.

B) E3 effects of filtering and abstention

Table 3 separates the two E2 components. Filter-only has the highest balanced score (0.9111); adding the original score-based abstention rule lowers it to 0.9048.

Table 2. Ablation on the test split (n = 210 per noise and incomplete condition, 70 contradictory).

Variant	Filter	Rule	Acc 0.0	Acc 0.4	Acc 0.8	Mean noise acc.	Rej inc.	Acc. contra.	Bal.
E0	no	no	0.948	0.971	0.719	0.879	0.852	0.543	0.8659
E2 filter only	yes	no	0.948	0.943	0.776	0.889	0.933	0.386	0.9111
E0 abstain only	no	yes	0.943	0.948	0.643	0.844	0.919	0.500	0.8817
E2 full	yes	yes	0.943	0.929	0.700	0.857	0.952	0.343	0.9048

The balanced-score effects are: filtering without abstention +0.0452, filtering with abstention +0.0230, abstention without filtering +0.0159, and abstention after filtering -0.0063.

Filtering improves both components of the score, but rejection dominates: 0.0405 of the 0.0452 gain (89%) comes from incomplete rejection rising from 0.852 to 0.933, noise accuracy contributes 0.0048. At noise_0.8, accuracy increases by 5.7 percentage points. Noise_0.0 is unchanged and noise_0.4 falls by 2.9 points.

The original abstention rule fires on 191 test contexts, including 139 incomplete questions. Of these, 156 filter-only predictions were already exact refusals, leaving 35 changed strings. The rule overrides 23 correct answers (20 noise and 3 contradictory), while gaining only 4 incomplete refusals. Full E2 therefore has lower mean noise accuracy than E0 (0.857 versus 0.879), its balanced-score improvement is driven by increased incomplete rejection, which outweighs that loss.

Retuning τ on calibration filter-only answers selects $\tau' = 1.08 \times 10^{-7}$, rather than disabling abstention. This raises the full-system test score from 0.9048 to 0.9079, but remains below filter-only (0.9111). It still

replaces 7 correct noise answers to gain 1 incomplete refusal. The best filter-only score among the swept test thresholds is 0.9111, equal to no added abstention, the best unfiltered score is 0.8849. These test-sweep results are descriptive, not a calibration procedure or proof about all possible thresholds.

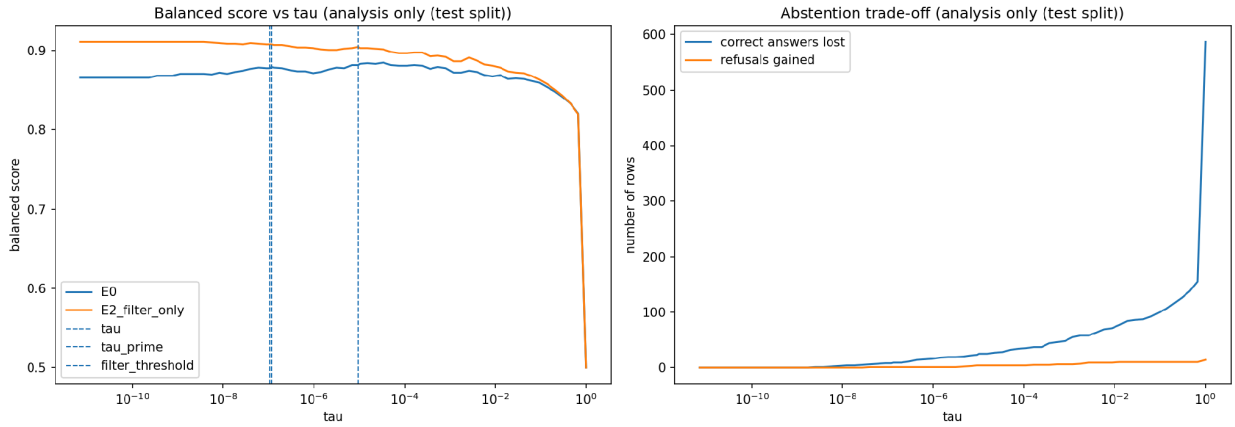


Figure 1. Test threshold sweep: balanced score for E0 and filter-only (left), and correct noise answers lost versus incomplete refusals gained after filtering (right). The tested thresholds do not improve filter-only beyond its value without score-based abstention.

Contradictory accuracy falls from 0.543 for E0 to 0.386 for filter-only, and 0.343 for full E2. Filtering loses 11 correct answers out of 70 relative to E0; full E2 loses 14. This trade-off is more substantial than the small aggregate-score differences and motivates the E4 analysis.

C) E4: where the remaining errors come from

Table 3 assigns each of the 146 full-E2 failures to one category.

Table 4. Failure categories of full E2 on the test split.

Category	Noise 0.0	Noise 0.4	Noise 0.8	Inc.	contra.	Total
refused despite evidence	7	6	28	–	28	69
wrong despite evidence	3	4	9	–	6	22
filter removed gold	1	2	9	–	9	21
abstention override	1	3	16	–	3	23
answered without evidence	–	–	–	10	–	10
gold not found	–	–	1	–	–	1
Total failures	**12**	**15**	**63**	**10**	**46**	**146**

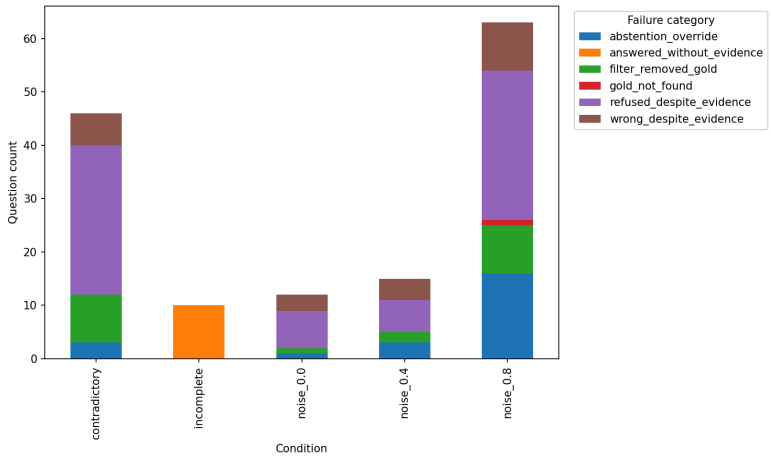


Figure 2. The same counts as a stacked bar chart by condition.

Most categorised failures occur despite retained evidence: 91 of 146 (62%) are refusals (69) or wrong answers (22) with a gold-bearing passage kept. Another 23 are correct answers overridden by abstention. Of 44 E0-correct questions lost after filtering, 34 still retain gold evidence. These cases implicate generation and refusal decisions as well as evidence selection.

Filtering removes all gold-bearing passages in 21 failures. It keeps only one passage in 16, including 7 fallback cases. The best gold passage ranks second in 7, third in 8, fourth in 4, and fifth in 2. Keeping at least the two highest-scoring passages would retain gold evidence in the 7 rank-2 cases, although its effect on answer accuracy requires testing.

All 23 abstention overrides discard a correct filter-only answer with gold evidence kept; 8 use the fallback. Low relevance scores therefore do not always imply that a supported answer is unavailable.

Full E2 answers 10 of 210 incomplete questions, compared with 31 for E0 and 14 for filter-only. None of these 10 uses the fallback or falls below τ: each has at least one negative passage scoring above both thresholds. Score-based abstention cannot reject these confidently mis-scored contexts.

The single gold_not_found case is a string-match miss: the gold title is "Dónde Vives", while the retained passage uses "Donde Vives". Full E2 refuses this question. The accent mismatch means the automatic evidence flag incorrectly treats the retained passage as lacking the gold string.

Table 5 groups the 70 contradictory contexts by retained passages. The true passage is removed in 9 cases. The fake passage is kept in 39, always together with the true passage.

Table 4. Full E2 labels on contradictory questions by passages kept.

Passages kept	n	Correct	Rejected	Wrong	Accuracy
True and fake	39	16	19	4	0.41
True only	22	8	12	2	0.36

Passages kept	n	Correct	Rejected	Wrong	Accuracy
Neither	9	0	6	3	0.00

Accuracy is 0.41 with both passages retained and 0.36 with only the true passage retained. Errors therefore persist after fake-passage removal, although these different question groups do not establish the causal effect of retaining fake evidence. Relevance filtering alone does not resolve the conflict condition. The substring metric is also lenient: 2 of the 24 correct predictions name the fake answer as well.

5.Discussion

Filtering improves the balanced score by 0.0452, with 89% of that gain coming from incomplete rejection. It also improves heavy-noise accuracy, but reduces contradictory accuracy. Neither calibration-selected abstention threshold improves on filter-only in the held-out test set.

The embedding-similarity baseline does not achieve this. E1's higher incomplete rejection is cancelled out by lost answerable accuracy, leaving its balanced score level with E0. E2 differs from E1 in both the relevance scorer and the number of passages kept, so this comparison cannot say which matters more; the E3 ablation only separates filtering from abstention within E2.

Full E2 raises incomplete rejection further but lowers mean noise accuracy below E0. A better balanced score can therefore coexist with worse answer accuracy; per-condition metrics and label transitions are needed to interpret the trade-off.

The 69 refusals and 22 wrong answers despite retained evidence, plus 23 abstention overrides, motivate improving generation and refusal decisions. The 21 gold-removal failures also motivate less aggressive filtering.

A) Novelty and contribution

Our contribution is a controlled investigation using the generator itself as a training-free relevance judge. A 2×2 ablation separates filtering from score-based abstention, and calibration-only retuning tests whether the original abstention threshold was mismatched to the filtered system. The passage-level failure taxonomy links aggregate metrics to retained evidence and specific decision failures. Each analysis notebook verifies upstream results, and the saved predictions allow E3 and E4 to be reproduced without model generation.

B) Limitations

- Small test set. Each noise condition has 210 test questions and the contradictory condition has 70. One question changes accuracy by 0.48 and 1.43 percentage points respectively. We report one

run without confidence intervals, so small differences should be interpreted cautiously; statistical significance has not been established.

- One model and one benchmark. All results use Qwen2.5-3B-Instruct as both scorer and generator, on RGB's English questions with RGB's passages. A larger model or a real retriever may behave differently.
- String-match evaluation. Substring matching misses paraphrases and accents and accepts answers that name both candidates. The evidence flags agree with RGB's labels for 99.8% of positives, but individual cases, such as the accent mismatch, can be misclassified.
- ● Quantized model. All results use the 4-bit Q4_K_M version of Qwen2.5-3B-Instruct served by Ollama; full-precision weights may score differently. Greedy decoding is not guaranteed to be bit-identical across machines or under parallel requests, E0 and E1 was generated on a different machine (TC1, one V100 GPU) from E2, so a few individual predictions could change on a rerun.
- Threshold transfer. The selected filter threshold is around 10^{-7} and the original abstention threshold around 10^{-5}. Their values are specific to the score distribution and may not transfer to a different model, precision, or prompt.

C) Future work

The failure analysis suggests specific changes. Replacing the single-passage fallback with a minimum of two kept passages targets the rank-2 filter losses. Changing the prompt or the refusal rule so that the model answers when a kept passage contains a candidate answer targets the largest category, refused despite evidence. For contradictory questions, detecting two kept passages that support different answers and either reporting the conflict or abstaining with a stated reason would address a failure that relevance filtering cannot. Finally, normalising accents and diacritics in the answer metric, and repeating the experiments with several seeds and bootstrap confidence intervals, would make the comparisons more precise.

6.Conclusion

On the baseline-aligned numeric-ID split, passage filtering improves the Ollama balanced score from 0.866 to 0.911, mainly through better rejection of unanswerable questions and additional gains under heavy noise. A fixed top-2 embedding-similarity filter with abstention (E1) leaves the balanced score essentially unchanged (0.868). Score-based abstention lowers held-out performance, including after calibration-only retuning. Contradictory accuracy worsens, and most categorised failures occur despite retained evidence. Future improvements should target evidence use, refusal decisions, and conflict handling. All methods share one quantized Ollama configuration, so these comparisons are not confounded by differences in inference setup.

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